

1. Language (recap)

- Propositional atoms: $\backslash(p, q, r, \dots\backslash)$, plus:
 - $\backslash(\mathsf{Model}e, \mathsf{Predict}e, \mathsf{Track}_e\backslash)$
 - $\backslash(\mathsf{SurpriseHigh}e, \mathsf{SurpriseLow}e, \mathsf{Notice}_e\backslash)$
- Modal operators (per entity $\backslash(e\backslash)$):
 - $\backslash(\Box^{\mathsf{Rec}}e, \Diamond^{\mathsf{Rec}}e\backslash)$
 - $\backslash(\Box^{\mathsf{Comp}}e, \Diamond^{\mathsf{Comp}}e\backslash)$
- Temporal operators (optional but useful):
 - $\backslash(P\varphi\backslash)$: “ $\backslash(\varphi\backslash)$ held at the previous moment”
 - $\backslash(F\varphi\backslash)$: “ $\backslash(\varphi\backslash)$ will hold at some future moment”

2. Base propositional system

Use any standard Hilbert system for propositional logic, e.g.:

- Axioms (all schemas):
 - P1: $\backslash(\varphi \rightarrow (\psi \rightarrow \varphi))\backslash$
 - P2: $\backslash((\varphi \rightarrow (\psi \rightarrow \chi)) \rightarrow ((\varphi \rightarrow \psi) \rightarrow \varphi \rightarrow \chi))\backslash$
 - P3: $\backslash((\neg \varphi \rightarrow \neg \psi) \rightarrow (\psi \rightarrow \varphi))\backslash$
- Rule:
 - Modus Ponens (MP): from $\backslash(\varphi\backslash)$ and $\backslash(\varphi \rightarrow \psi\backslash)$, infer $\backslash(\psi\backslash)$.

3. Modal part (normal K for each operator)

For each entity $\backslash(e\backslash)$, for each $\backslash(\Box^{\mathsf{Rec}}e\backslash)$ and $\backslash(\Box^{\mathsf{Comp}}e\backslash)$:

- Axiom K (distribution):
 - K-Rec($_e\backslash$):
$$\backslash[\Box^{\mathsf{Rec}}e(\varphi \rightarrow \psi) \rightarrow (\Box^{\mathsf{Rec}}e \varphi \rightarrow \Box^{\mathsf{Rec}}e \psi)\backslash$$
 - K-Comp($_e\backslash$):
$$\backslash[\Box^{\mathsf{Comp}}e(\varphi \rightarrow \psi) \rightarrow (\Box^{\mathsf{Comp}}e \varphi \rightarrow \Box^{\mathsf{Comp}}e \psi)\backslash$$

$\Box^{\{\text{Comp}\}}e(\varphi \rightarrow \psi) \rightarrow \Box^{\{\text{Comp}\}}e \varphi$
 $\rightarrow \Box^{\{\text{Comp}\}}_e \psi$
 $\}$

- Necessitation rules:

- Nec-Rec(e): from $\vdash \varphi$, infer $\vdash \Box^{\{\text{Rec}\}}e \varphi$
- Nec-Comp(e): from $\vdash \varphi$, infer $\vdash \Box^{\{\text{Comp}\}}e \varphi$

(That makes both modalities normal.)

You can add frame-specific axioms (T, 4, 5, etc.) if you want reflexive, transitive, or Euclidean behavior, but they're not required for the recognition/comprehension distinction itself.

4. Temporal part (minimal)

If you want temporal reasoning, add:

- Axioms for $\Box(F)$ (future, LTL-style, minimal):

- F1: $\Box(F\varphi) \leftrightarrow (\varphi \vee X\Box\varphi)$ if you had a next operator X ;
- but since we only need “eventually” schematically, we can treat $\Box(F)$ as a primitive with:
- Monotonicity: $(\varphi \rightarrow \psi) \rightarrow (\Box(F\varphi) \rightarrow \Box(F\psi))$

- Axioms for $\Box(P)$ (previous):

- Monotonicity: $(\varphi \rightarrow \psi) \rightarrow (\Box(P\varphi) \rightarrow \Box(P\psi))$

- Rules:

- From $\vdash \varphi \rightarrow \psi$, infer $\vdash \Box(F\varphi) \rightarrow \Box(F\psi)$
- From $\vdash \varphi \rightarrow \psi$, infer $\vdash \Box(P\varphi) \rightarrow \Box(P\psi)$

(We're keeping temporal structure intentionally light—just enough to express your “just before” invariant.)

5. Non-logical axioms encoding recognition vs comprehension

These are domain-specific axiom schemas in the same proof system.

5.1 Comprehension constraints

For each entity e :

1. Model requirement

$$\begin{array}{l} \backslash[\\ \textbf{C1}e:\text{quad } \Box^{\text{Comp}}e \top \rightarrow \text{Model}_e \\ \backslash] \end{array}$$

2. Predictive power

$$\begin{array}{l} \backslash[\\ \textbf{C2}e:\text{quad } \Box^{\text{Comp}}e \top \rightarrow \text{Predict}_e \\ \backslash] \end{array}$$

3. Continuous tracking

$$\begin{array}{l} \backslash[\\ \textbf{C3}e:\text{quad } \Box^{\text{Comp}}e \top \rightarrow \text{Track}_e \\ \backslash] \end{array}$$

4. Low surprise

$$\begin{array}{l} \backslash[\\ \textbf{C4}e:\text{quad } \Box^{\text{Comp}}e \top \rightarrow \text{SurpriseLow}_e \\ \backslash] \end{array}$$

5.2 Recognition constraints

1. Recognition lacks full model or prediction

$$\begin{array}{l} \backslash[\\ \textbf{R1}e:\text{quad } \Box^{\text{Rec}}e \top \rightarrow (\neg \text{Model}_e \vee \neg \\ \text{Predict}_e) \\ \backslash] \end{array}$$

2. Recognition tied to high surprise or lack of tracking

$$\begin{array}{l} \backslash[\\ \textbf{R2}e:\text{quad } \Box^{\text{Rec}}e \top \rightarrow (\text{SurpriseHigh}_e \vee \neg \\ \text{Track}_e) \\ \backslash] \end{array}$$

5.3 Mutual incompatibility (optional but natural)

3. No simultaneous pure comprehension and recognition

$$\begin{array}{l} \backslash[\\ \textbf{M1}e:\text{quad } \Box^{\text{Comp}}e \top \rightarrow \neg \Box^{\text{Rec}}_e \top \\ \backslash] \end{array}$$

(You can also add the converse if you want strict exclusivity.)

6. Noticing invariant as a theorem schema

We encode your key idea:

> If I notice $\neg(e)$ now, then just before now I was not in comprehension state for $\neg(e)$.

Add as a non-logical axiom schema:

- N1 (Noticing $\Rightarrow$ high surprise):

$$\begin{array}{l} \backslash[\\ \text{\texttt{\textsf{Notice}}}e \rightarrow \text{\texttt{\textsf{SurpriseHigh}}}e \\ \backslash] \end{array}$$

- N2 (Noticing $\Rightarrow$ no prior comprehension):

$$\begin{array}{l} \backslash[\\ \text{\texttt{\textsf{Notice}}}e \rightarrow P(\neg \Box^{\text{\texttt{\textsf{Comp}}}}e \top) \\ \backslash] \end{array}$$

These are now available for derivations like any other axiom.

7. Example derivation sketch

Goal: derive that if comprehension holds, noticing cannot occur at the same moment (for the same $\neg(e)$), under a natural extra assumption.

Add one more domain axiom:

- Link surprise to noticing (no noticing under low surprise):

$$\begin{array}{l} \backslash[\\ \text{\texttt{\textbf{S1}}}e : \text{\texttt{\textsf{SurpriseLow}}}e \rightarrow \neg \text{\texttt{\textsf{Notice}}}_e \\ \backslash] \end{array}$$

Now:

1. From $C4\neg(e)$: $\neg(\Box^{\text{\texttt{\textsf{Comp}}}}e \top) \rightarrow \text{\texttt{\textsf{SurpriseLow}}}_e$.

2. From $S1\neg(e)$: $\text{\texttt{\textsf{SurpriseLow}}}e \rightarrow \neg \text{\texttt{\textsf{Notice}}}_e$.

3. By propositional reasoning (P2, P1, MP), derive:

$$\begin{array}{l} \backslash[\\ \Box^{\text{\texttt{\textsf{Comp}}}}e \top \rightarrow \neg \text{\texttt{\textsf{Notice}}}e \\ \backslash] \end{array}$$

i.e. in comprehension mode, noticing of $\neg(e)$ cannot occur—it's already smoothly tracked.

That matches your intuition: noticing is a marker of non-comprehension or model failure, not of stable understanding.